Introduction to Finance for Data Scientists

Session 8: Scoring

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Algorithmic Scoring and Pricing

- Credit markets

  mortgages, consumer loans, business loans

  $\Rightarrow$ lender must predict default

- Insurance markets

  health, property and casualty

  $\Rightarrow$ insurer must predict losses and damages

- Accurate prediction is key

  price too low $\Rightarrow$ lose money

  price too high $\Rightarrow$ lose market shares

Algorithmic Scoring and Pricing

- Important challenges your studied in the course "Interpretability, Stability, and Algorithmic Fairness"

- Today: strategic behavior of clients and competitors $\Rightarrow$ additional challenges

Road Map

Mini-Case: Zillow Offers

Adverse Selection

Lending Game

Hirshleifer Effect

Mini-Case: Upstart

Zillow Offers

- Zillow founded in 2006 in Seattle by former Expedia executives
- Initial core business: digital real-estate platform
  - Home search, listings, rentals, housing information
  - Revenue from advertising, lead generation, and adjacent services (mortgages)
- In 2018 Zillow launched **Zillow Offers** = "iBuyer" service
  - Homeowners request an offer online from Zillow
  - Zillow evaluates the home with its algorithm and makes an offer
  - If accepted, Zillow buys the home and handles closing
  - Zillow performs light repairs
  - Zillow relists and resells the home

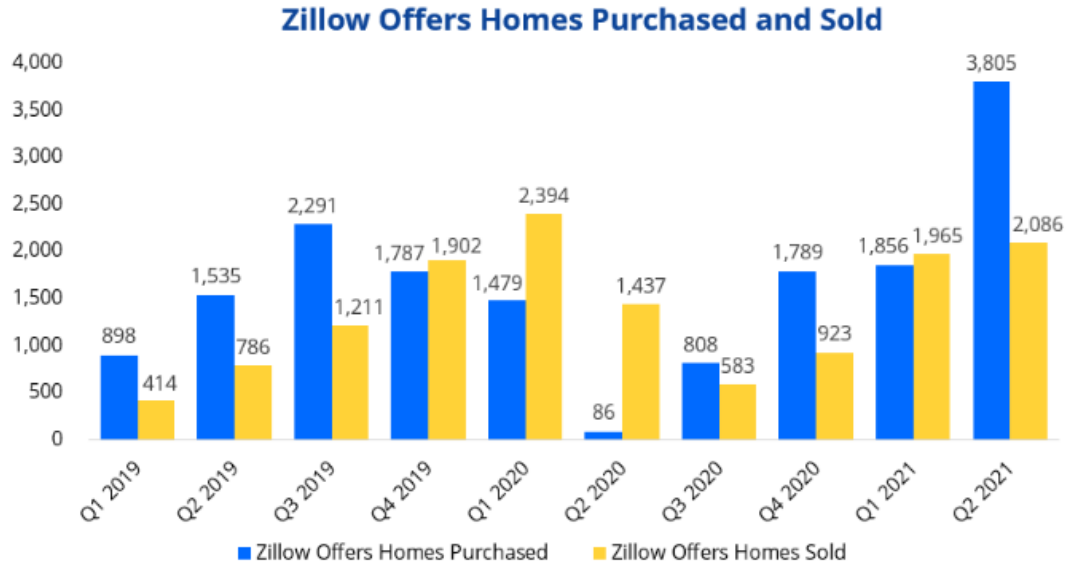
$\Rightarrow$ Zillow moved from platform intermediary to principal buyer and seller

- Value proposition to sellers: speed, certainty, and convenience (avoid repeated showings)

Estimating Home Values

- Data used
  - Home characteristics: square footage, location, photos, etc.
  - On-market data: comparable homes' listing price, days on market
  - Off-market data: tax assessments, prior sales, public records
  - Market trends: local price dynamics and seasonal demand
- Model
  - Neural network model
  - From $1 million Kaggle competition on home valuation prediction
  - Input: see above
  - Output: estimated market value of the home
- Model performance (Apr 2021): https://www.zillow.com/zestimate/

Financial Performance



Second-Quarter 2021 Zillow Offers Unit Economics

	Total	Average per Home	% of Revenue
Homes sold	2,086		
Zillow Offers revenue	$ 772,030,000	$ 370,101	100.0%
Operating costs:			
Home acquisition costs*	672,592,000	322,432	87.1%
Renovation costs*	20,862,000	10,001	2.7%
Holding costs*,**	4,527,000	2,170	0.6%
Selling costs	29,544,000	14,163	3.8%
Total operating costs	727,525,000	348,766	94.2%
Return on Homes Sold Before Interest Expense[3]	44,505,000	21,335	576 bps
Interest expense*,**	3,545,000	1,699	0.5%
Return on Homes Sold After Interest Expense[3]	$ 40,960,000	$ 19,636	531 bps

Zillow Offers EBITDA 2021Q2: negative $29 million (negative 4% of revenue)

Q1. Why is gross margin positive but EBITDA negative?

Q2. How to increase EBIDTA?

Financial Performance

- EBITDA 2021Q3: negative $380 million (negative 32% of revenue)

- Shareholder letter 2021Q3:

  "We recorded a $304 million write-down on inventory owned at the end of Q3 as a result of unintentionally purchasing homes at higher prices than our current estimates of future selling prices."

  "We have been able to convert only about 10% of the serious sellers who ask for a Zillow Offer."

Q. What is the connection between the inventory write-down and the 10% seller conversion rate?

Winner's Curse

- Unbiased estimate $\Rightarrow$ estimate is above the fair value as often as it is below the fair value
- The potential seller is more likely to accept Zillow's offer if it is above fair value, i.e., when Zillow is overpaying
- This is the **winner's curse** (or **adverse selection**)

How to Mitigate Adverse Selection?

1. Bid more conservatively than the statistical estimate
   - Create a margin of safety against overpaying
   - But lower conversion rate
2. Backtest after launch
   - Compare forecast values with realized resale outcomes
   - But reactive: losses may already be incurred
3. Experiment with pricing on a small sample, then backtest
   - Learn how offer levels affect acceptance, resale margins, and selection
   - But slower scaling
4. Focus on easy-to-value, liquid homes and markets
   - Reduce hidden information and pricing error
   - But weaker value proposition and narrower market

Winner's Curse Everywhere

- **Lending:** Borrowers accept your loan offer when other lenders charge higher rates, indicating your credit risk estimate is too optimistic
- **Insurance:** Customers buy insurance from you when other insurers charge higher premiums, indicating your risk estimate is too optimistic
- **Trading:** Glosten-Milgrom model of market making studied with Jean-Edouard
- Auctions, etc.

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Statistical Underpinnings of Adverse Selection

- Highly relevant for the group work
- You make loans to businesses
- Offer interest rate R, paid at issuance (to simplify formulas)
- Cash flow (per euro lent) if loan offer is accepted:

  at issuance: $-1+R$

  at maturity: $\begin{cases}1 \text{ if no default} \\ 0 \text{ if default}\end{cases} = 1-D$ where $D \in \{0,1\}$ is default indicator

- Assume zero discount rate. Profit $= (-1+R)+(1-D) = R-D$
- Expected profit: NPV $= R - \mathbb{E}(D) = R - \mathbb{P}[Default]$

Statistical Underpinnings of Adverse Selection

- Default probability: $\mathbb{P}[Default] = f(X) + U$
  - X: borrower characteristics observed by lender
  - U: unobserved determinants of default risk ($U \perp X$, $\mathbb{E}[U] = 0$)
- Lender's best estimate of default probability: $f(X)$
- Suppose each loan applicant offered an interest rate R takes the loan offer if R is below a threshold: $R < \overline{R} + \mathbb{P}[Default]$, where the threshold is higher for riskier applicants
  - Rationale: Riskier applicants are more likely to accept a given offer because their chances to get a loan from another lender are lower
  - $\overline{R}$ is the maximum rate a risk-free borrower would accept
- How should the lender set the interest rate? → Simple idea: take a margin over estimated default probability: $R = f(X) + M$ with $M > 0$

Q. What is the NPV per euro lent (i.e., on accepted loan offers)?

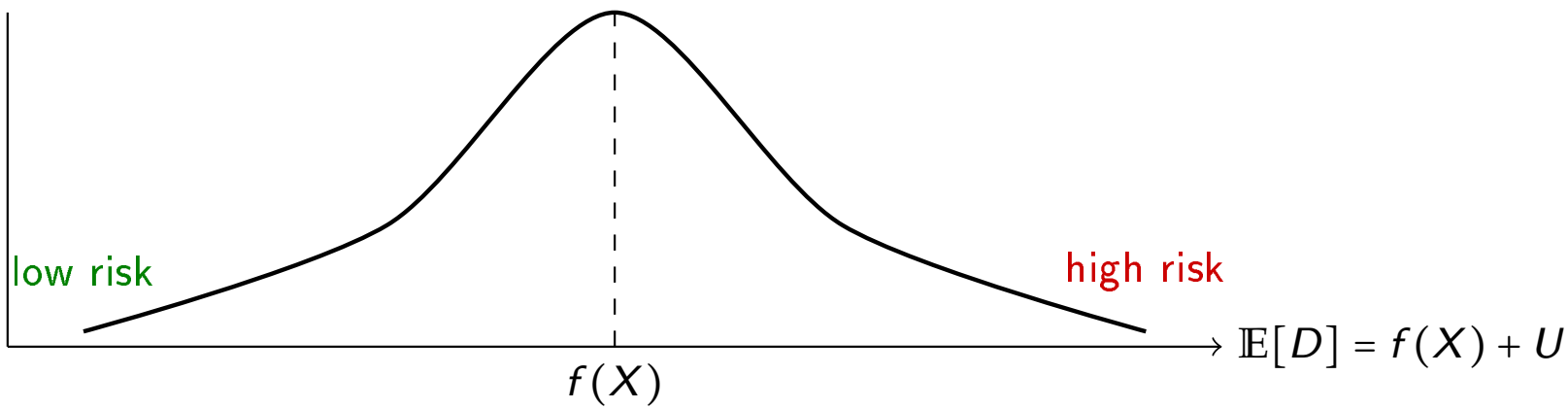
a. M b. more than M c. less than M d. it depends

Statistical Underpinnings of Adverse Selection

- NPV per euro lent on accepted loan offers

$$
\begin{aligned}
&= R - \mathbb{E}\big[D \,|\, \text{loan offer is accepted}\big] \\
&= R - \mathbb{E}\big[D \,|\, R < \overline{R} + f(X) + U\big] \\
&= f(X) + M - \mathbb{E}\big[f(X) + U \,|\, f(X) + M < \overline{R} + f(X) + U\big] \\
&= M - \mathbb{E}\big[U \,|\, U > M - \overline{R}\big] \\
&< M
\end{aligned}
$$

$\Rightarrow$ Adverse selection: risky applicants are more likely to accept the loan offer

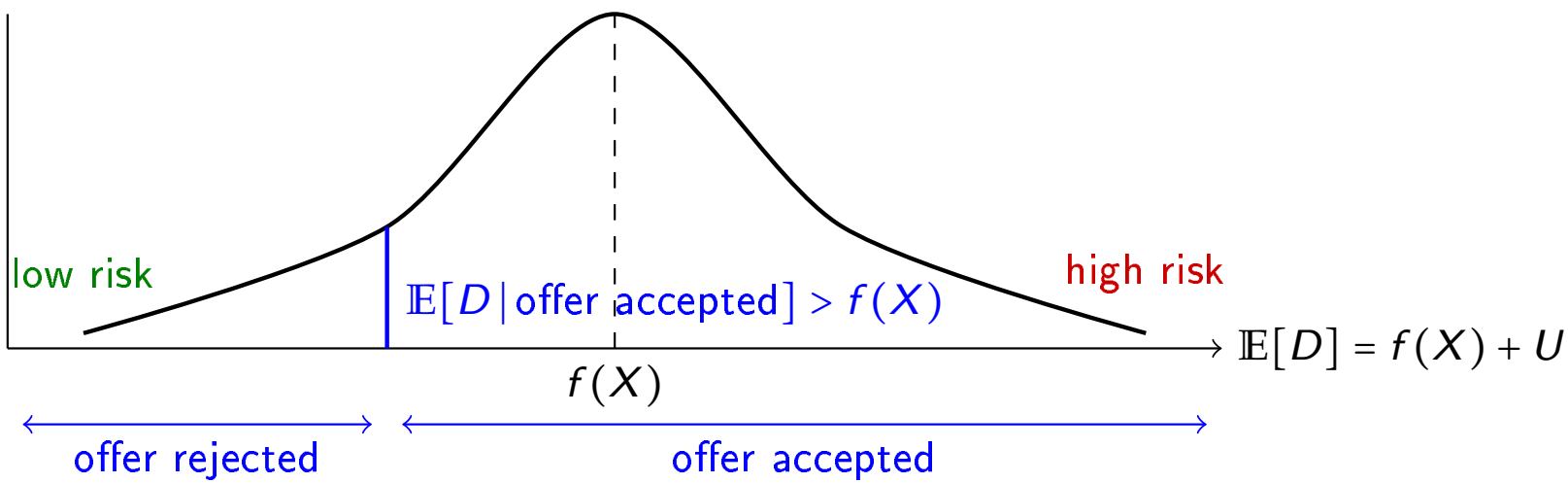


Statistical Underpinnings of Adverse Selection

- NPV per euro lent on accepted loan offers

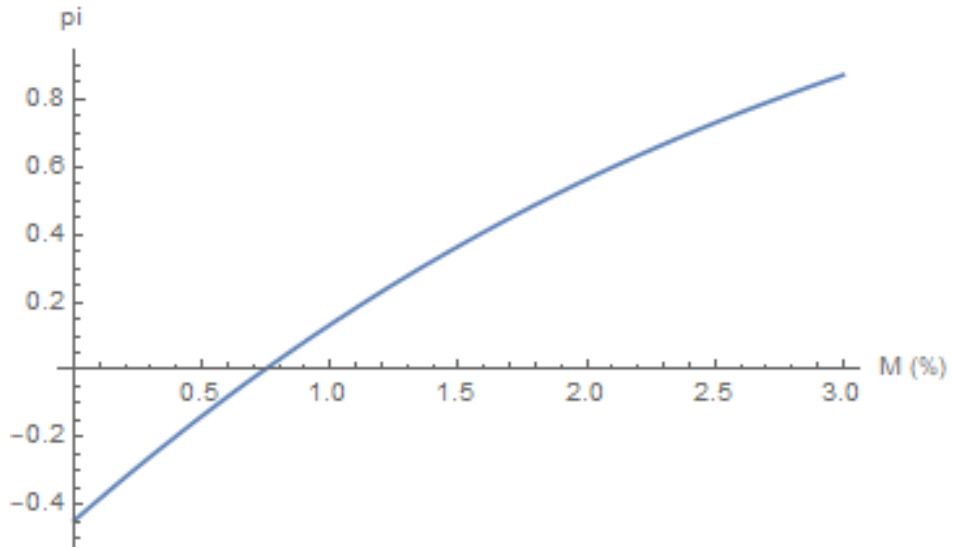
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\end{aligned}
$$

$\Rightarrow$ Adverse selection: risky applicants are more likely to accept the loan offer



Statistical Underpinnings of Adverse Selection

- Expected profit per euro lent as function of M



$\overline{R} = 2\%$; $U \sim \mathcal{N}(0, 1.8\%)$

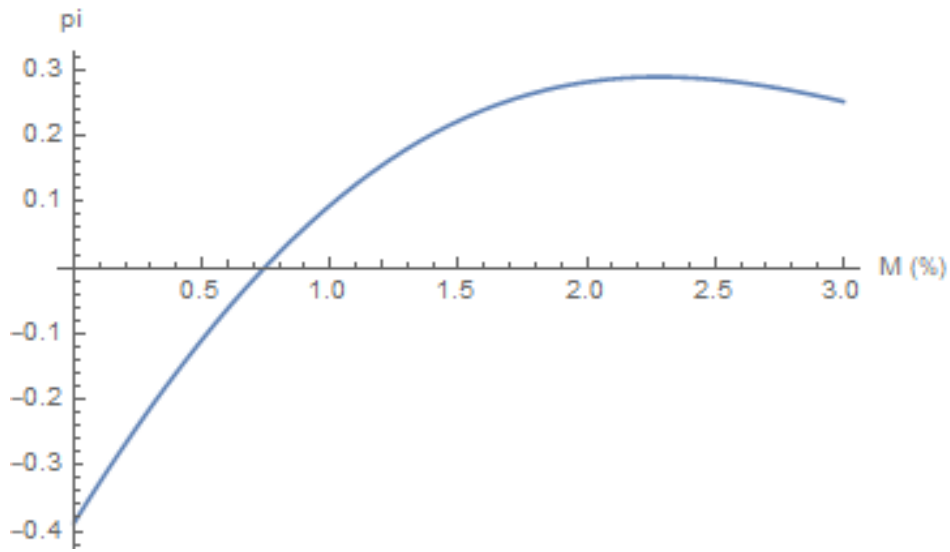
- Adverse selection $\Rightarrow$ profit is less than M; can be negative even if $M > 0$

- Next question: Which M maximizes expected profit?

Statistical Underpinnings of Adverse Selection

- Expected profit

$$= E[(R-D)\times 1\{\text{offer accepted}\}] = \underbrace{E[R-D\,|\,\text{offer accepted}]}_{\uparrow M \text{ (previous slide)}} \times \underbrace{P[\text{offer accepted}]}_{\downarrow M}$$



- Why hump-shaped? Higher $M \Rightarrow$ higher profit per offer accepted but fewer offers are accepted

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Hirshleifer Effect

Mini-Case: Upstart

Group Work: Lending Game

- 3 fintech lenders (=3 teams of students) compete to make loans
- Data on past loans: predictors + default
- New loan applications
- All teams have data on the same past loans and receive the same new loan applications, but each team has different predictors

Stage 0: NOW Form teams of 4±1 persons and email your team's composition today to hombert@hec.fr

Stage 1: Each team makes an offer to every loan applicant → offers due on October 17

Each loan applicant chooses which team's loan offer to take. Applicants repay or default → profits and losses for each team

Stage 2: Each team improves its strategy based on the experience of stage 1 and play the same game again → offers + report due on October 23

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Information in Insurance

- Suppose we discover how to predict perfectly who will get sick, but this foreknowledge does not help to prevent or cure diseases

- Insurers use this information to price health insurance

Q1. Will this make people better or less-well insured?

a. better insured b. less-well insured

Q2. Will this make insurers more or less profitable?

a. more profitable b. less profitable

Information in Insurance

- What will happen?
  - No insurers accept to insure people predicted to be sick
  - People predicted to be healthy don't need insurance

$\Rightarrow$ The health insurance market breaks down

  - People are worse off: they can't get insurance
  - Insurers are worse off: they can't sell insurance

- **Hirshleifer effect:** Information can destroy insurance

How to Overcome the Hirshleifer Effect?

- Suppose an insurer announces it will not use the information and the promise is credible

Q3. Does this overcome the Hirshleifer effect and allow the insurer to sell insurance?

a. yes b. no

How to Overcome the Hirshleifer Effect?

- Suppose an insurer announces it will not use the information and the promise is credible

Q3. Does this overcome the Hirshleifer effect and allow the insurer to sell insurance?

a. yes b. no

- No, because of adverse selection
  - People predicted to be healthy are offered cheap insurance from other insurers, or they don't even buy insurance
  - The insurer only gets people who will be sick, so it cannot insure them

- See practice problem 2

How To Overcome the Hirshleifer Effect?

- Solution 1: Regulation
  - Forbid the use of some information (e.g., insurers are not allowed to use DNA testing)
- Solution 2: Government-provided insurance
  - Mandatory pooling (e.g., CatNat in France)

Insurance and Repricing

- Insurance contracts are exposed to the risk of information arrival and repricing

What happens to insurance premiums after the LA wildfires?

Longstanding insurance issues in the Golden State are sure to be exacerbated by the fires roaring through the LA area

January 8, 2025, 3:55 pm *By Sarah Wolak*



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Upstart

- Founded in 2012 in California
- Initial business: personal loans to college graduates paid back through income sharing agreements (marketing slogan: "The startup is *you*")
- In 2014 it pivoted to traditional personal loans, using education variables (college attended, field, GPA) as underwriting criteria in additional to traditional variables

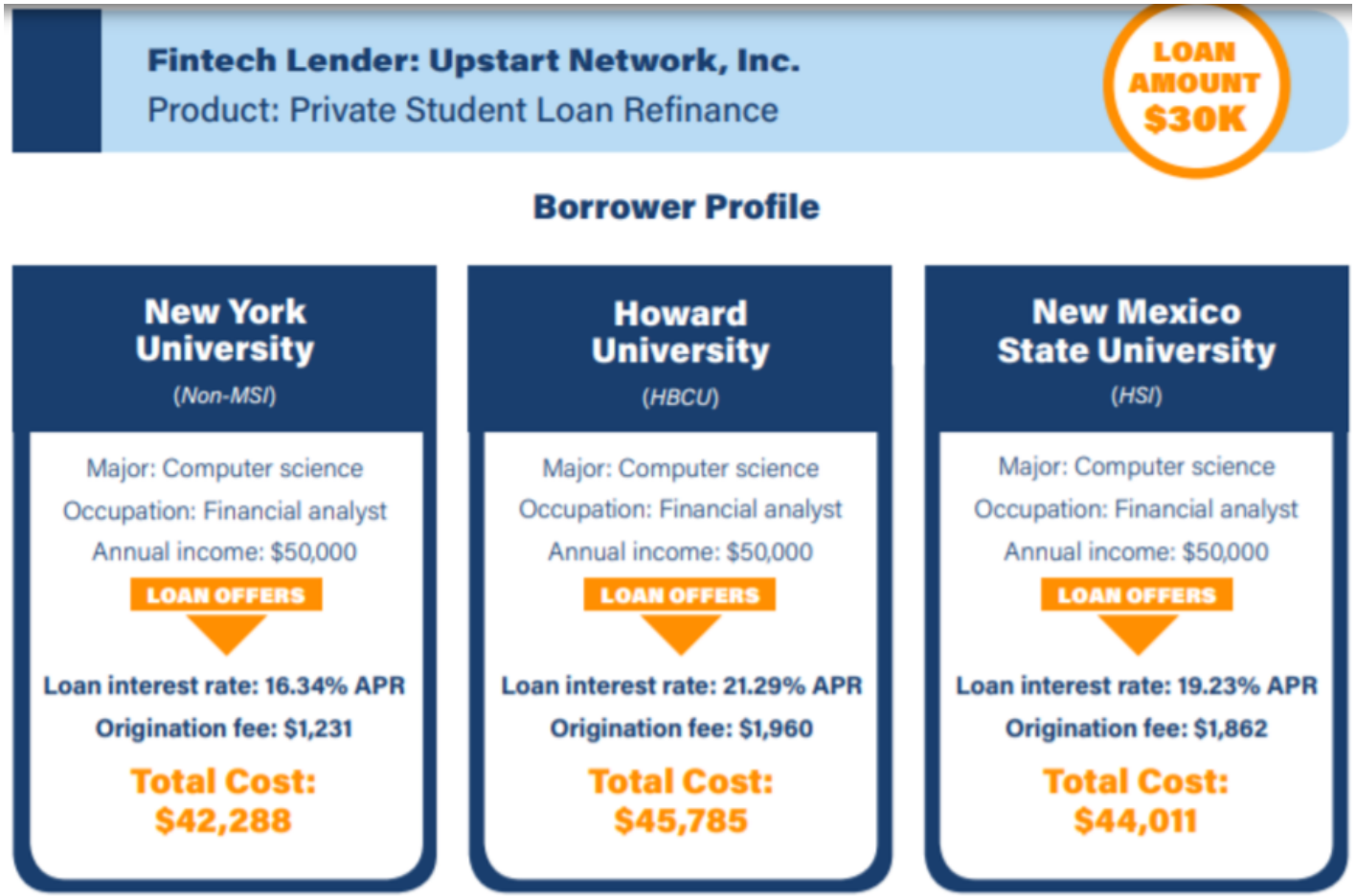
Upstart

Wells Fargo, Upstart criticized after study finds loan disparities

Published Feb. 6, 2020 • Updated Feb. 14, 2020

The request comes a week after the nonprofit Student Borrower Protection Center found that an Upstart borrower who attended historically black Howard University would pay thousands of dollars more on average for a five-year loan than a borrower with an identical credit profile who studied at New York University.

Upstart



Q. What may explain the discrimination?

Algorithmic Fairness

- College attended predicts default
- By screening on college attended, the algorithm's prediction is (unintendedly) correlated with race
- Solution: algorithm interpretability → see the course "Interpretability, Stability, and Algorithmic Fairness"

Invisible Primes: Fintech Lending with Alternative Data*

Marco Di Maggio† Dimuthu Ratnadiwakara‡

January 4, 2024

Abstract

We use anonymized data and an underwriting algorithm that incorporates alternative data from a major platform to study how alternative data affects credit access and borrower outcomes. Comparing actual outcomes of the fintech platform's model to counterfactual outcomes based on a "traditional model" used for regulatory reporting purposes, we find that the fintech platform's model approves 15-30% of low credit score applicants rejected by the traditional model and offers substantial reductions in interest rates. The borrowers most positively affected are the "invisible primes"—borrowers with low credit scores and short credit histories, but also a low propensity to default. Some high credit score borrowers are instead rejected by the platform. About two-thirds of the effects are due to the inclusion of additional data, while the remainder is due to a more sophisticated underwriting model. Leveraging exogenous variations in credit access, we show that funding loans to invisible primes leads to better economic outcomes for borrowers.

6th CFPB Research Conference, 5th Annual IMF Macro-Financial Research Conference, Eighth ECB Annual Research Conference 2023, and 4th Future of Financial Information Conference for helpful comments. We thank Upstart Network, Inc. for providing access to its data. Special thanks to Don Carmichael for his assistance in setting up access to the data infrastructure on the cloud and for his assistance in navigating

Redistributive Effects

- More information creates winners and losers
  - Winners: those for whom additional information predicts low risk (e.g., went to NYU)
  - Losers: those for whom additional information predicts high risk (e.g., went to minority-attended college)

Conclusion

- Email me the composition of your team for the lending game
- Practice problem on Slack
- I hope you enjoyed the course and learnt a lot 